

PRATIK PATIL

Senior Software Engineer | Full Stack Engineer | AI Product UI Engineer

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Authorized to work in US (H1B Transfer required)

PROFESSIONAL SUMMARY

Senior frontend engineer with 6+ years building production React web and React Native mobile applications. At BDIPlus, delivered six client and product workstreams for American Express, Citizens Bank, Morgan Stanley, and Afficiency under a Microsoft distribution partnership, spanning OneDATA.Plus customer-data platform, ChatCDP.ai LLM chat product, and ModenX in-store retail mobile app. Led 28-module React 18 migration, cut Time to Interactive from 7.2s to 2s, engineered a 10K-node D3.js network graph for marketing intelligence, and built an LLM chat UI with Server-Sent Events streaming and embedded Apache Superset analytics. M.S. Computer Science, Stevens Institute of Technology. Microsoft Azure (AZ-900) certified.

CORE TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, Java, SQL, PL/SQL, HTML5, CSS3

Frontend: React 18, React Hooks, Next.js, Redux Toolkit, React Router v6, React Native, React Flow, D3.js, Tailwind CSS, Material UI, shadcn/ui, Responsive Design, Mobile-First Development

Backend and APIs: Node.js, Express.js, REST APIs, GraphQL, Server-Sent Events (SSE), Socket.IO, Stripe Checkout, Webhooks, OAuth 2.0, SSO, SAML, Multi-Tenant Architecture

AI and LLM: LLM-Powered Product UIs, SSE Token Streaming, Structured Output Rendering, Embedded Analytics with Apache Superset, Vercel AI SDK, Cursor, Claude Code

Databases: PostgreSQL, MongoDB, Redis, Firebase, Oracle 11g, MySQL

Cloud and DevOps: AWS (S3, Lambda, CloudFront), Microsoft Azure (AZ-900 Certified), Vercel, GitHub Actions, CI/CD, Git

Testing: Jest, React Testing Library, Vitest, Playwright

Python and Data: Pandas, NumPy, Selenium, BeautifulSoup, Flask, NLTK, Tkinter, Matplotlib, Jupyter Notebook

BI and ETL: Apache Superset, Tableau, Talend ETL, PL/SQL Reporting, Microsoft Excel

Tools and Workflow: Webpack, Lighthouse, Chrome DevTools, Postman, Jira, Figma, Agile and Scrum

PROFESSIONAL EXPERIENCE

Senior Software Engineer, Full Stack | BDIPlusNew York, NY | Jul 2022 to Apr 2026

Enterprise SaaS for banking compliance and customer data platforms. Clients include American Express, Citizens Bank, Morgan Stanley, and Afficiency; Microsoft as distribution partner.

American Express : Real-Time Workflow Management Tool

- Built real-time workflow management tool for American Express in React and React Flow, visualizing in-flight customer-data workflows, owners, stage transitions, and live status as backend events streamed in.
- Implemented drag-and-drop reordering, status updates, and zoom-and-pan navigation so operations leads could manage dense workflows in one surface.
- Wired role-based access so only authorized owners could advance tasks, and surfaced inline audit trails on each node for compliance review.
- Integrated workflow surface with broader BDIPlus customer-data platform via REST and GraphQL APIs so state stayed in sync with downstream pipelines and dashboards.

OneDATA.Plus (ODP) : Core SaaS Customer-Data Platform

- Engineered OneDATA.Plus core network graph in D3.js, rendering up to 10K live entities concurrently as a star-field universe view; node size scales with connection density.
- Backed graph with real-time data fetch, virtualized rendering, zoom-and-pan navigation, and on-demand streaming from S3, keeping UI responsive on large underlying datasets.
- Surfaced highest-degree entities visually so marketing teams at American Express and similar clients could spot most-connected customers as priority outbound-campaign targets.
- Led 28-module React 18 migration in 7-day sprint; restructured codebase into self-contained feature modules with config-driven routing and barrel exports.

- Reorganized platform sidebar into six hub sections via single navigationConfig.js, letting product ship new modules without touching routing code.
- Wired React Router v6 with React.lazy and Suspense for per-route code splitting, cutting initial bundle size and removing duplicated chunks across modules.
- Cut Time to Interactive from 7.2s to 2s via React.lazy code splitting, React.memo and useMemo on data-heavy components, and dependency pruning; verified in Lighthouse and Chrome DevTools.
- Implemented multi-tenant SaaS billing across four pricing tiers using Stripe Checkout, SSO, OAuth 2.0, and RBAC; co-designed webhook flow for upgrade, downgrade, and dunning events.
- Instrumented banking-compliant self-hosted error logging with React Error Boundaries and authored runbook documentation, replacing SaaS vendor blocked by compliance review.
- Delivered reusable component library extending Material UI (pagination, skeleton loaders, virtualized grid, permission-gated views, modal system) adopted across six modules to standardize UX.
- Refactored legacy surfaces during React 18 migration, eliminating dead code; halved local compile time and shortened CI feedback loop on owned modules.

ModenX : In-Store Retail Mobile Experience (Retailer + Customer)

- Shipped ModenX in-store retail React Native app to iOS App Store and Google Play in 3 months; owned full lifecycle from spec through code signing and store submission.
- Designed two parallel surfaces inside one app: retailer view for in-store staff to manage inventory, offers, and customer interactions, and customer view for shoppers to browse, scan, and check out.
- Built native module integrations, push notification flow, deep linking, and offline-first state handling so app stayed functional through spotty in-store connectivity.
- Coordinated 5-engineer team through code reviews, weekly sprint reviews, and architecture decisions; established mobile coding conventions and folder structure adopted as team standard.
- Diagnosed and resolved memory and OS-compatibility issues across Android device matrix to stabilize production rollout.

Citizens Bank : Marketplace UI

- Delivered full marketplace UI for Citizens Bank in 3 months under critical client deadline, owning surface end to end from architecture through production launch.
- Designed responsive product listing, filtering, search, and detail flows in React with shared component library; implemented pagination, skeleton loaders, and permission-gated views.
- Wired role-based admin configuration with multi-tenant theming, letting marketplace be reskinned per client business unit without code changes.
- Hardened surface against compliance review with self-hosted logging, audit trails on user actions, and explicit error states for external dependencies.

ChatCDP.ai : LLM-Powered Customer-Data Chat Interface

- Architected ChatCDP.ai chat surface in React 18 and Redux Toolkit with SSE streaming; used by Morgan Stanley analysts to query the customer-data platform in natural language.
- Embedded Apache Superset dashboards directly inside LLM chat UI, letting analysts pivot from natural-language responses into interactive BI charts without leaving conversation.
- Designed structured-output rendering for JSON payloads, dynamic tables, and summary cards inside chat surface, removing need for analysts to parse raw model output.
- Co-designed SSE event contract with backend (status, partial, complete, and error event types) and added client-side handling to keep UI responsive during heavy streaming.
- Built reusable LLM UI primitives covering streaming token renderer, retry, cancel, error recovery, and conversation memory panel for use across chat product.
- Wired follow-up suggestions, conversation export, and prompt-history navigation, shaping surface around how analysts work rather than generic chatbot.

Afficiency : Dynamic Health Insurance Product for California State

- Adapted Afficiency's health insurance product for California rollout by building dynamic UI layer that adjusts forms, validations, and product variants from regulatory documentation.
- Designed configuration-driven rendering for fields, conditional logic, eligibility rules, and disclosure copy so product surface could be updated through config rather than code releases.

- Translated raw regulatory and client requirement documentation into structured product schema, working directly with client to align on edge cases and compliance constraints.
- Hardened rendering layer with input validation, error fallbacks, and consistent error states across surface to meet Afficiency's launch quality bar.

Engineering Practice and Mentorship

- Authored Jest and React Testing Library suites against critical paths across owned modules, raising coverage and cutting recurring production incidents through earlier defect detection.
- Mentored 3 junior engineers through pair programming, structured code reviews, and onboarding documentation; ran technical demos for stakeholders during bi-weekly sprint reviews.
- Translated migration trade-offs and performance constraints into language non-engineers could act on across product, design, and compliance reviews.

Application Development Analyst | AccenturePune, India | Oct 2018 to Jul 2021

Global delivery center; data engineering and Python automation for Fortune-500 client.

- Validated, deduplicated, and geocoded 1.2M customer site addresses across 195 countries using Python (Pandas, NumPy, Selenium, BeautifulSoup), delivering 4x throughput over legacy manual workflow.
- Improved overall data-quality solution by 96%, reducing rework cycles and shrinking ETA on client deliverables.
- Aided in securing client bids worth £330K GBP through faster, cleaner data delivery against tight bid timelines.
- Owned Python tooling as SME; built reusable scripts for scraping, validation, deduplication, and reporting across global research-park sites.
- Engineered ETL and reporting pipelines combining Python, PL/SQL, Talend ETL, and Tableau for client-facing analytics dashboards.
- Recognized with Apex Award for Innovation and Thought Leadership and BT Spark Award for Standard Performer.

SELECTED PROJECTS

Bio Maker : Marriage Biodata Generator | [Live Demo](#)

- Launched marriage biodata generation web app for Indian families and NRI diaspora using Next.js 15, TypeScript, Tailwind CSS v4, and shadcn/ui with mobile-first responsive forms and sub-2-second cold load.
- Integrated client-side PDF export with jsPDF and html2canvas, Vercel serverless functions for stateless generation, and PostHog for product analytics.
- Drove product end to end from strategy through information architecture, build, deployment, and growth analytics; freemium product-led growth model with one-time premium purchases.

TULSEE : Real-Time Work Management and Collaboration Platform | [GitHub](#)

- Composed B2C real-time collaboration platform in React, Redux, Next.js, Chakra UI, Node.js, Express, Firebase, and Socket.IO with personal and group projects, task CRUD, archive, and full-text search.
- Added Zoom API integration for in-platform meetings, group chat with end-to-end encryption, image and document upload, and Google SSO sign-in.

EDUCATION AND CERTIFICATIONS

M.S. Computer Science, Stevens Institute of Technology, Hoboken, NJ Aug 2021 to Dec 2022

GPA: 3.8 / 4.0

B.E. Information Technology, University of Pune, Pune, India Aug 2014 to Jun 2018

GPA: 3.7 / 4.0

- Microsoft Certified: Azure Fundamentals (AZ-900), Microsoft, Jan 2021